

AI-Powered Waste Management: Predictive Modeling for Sustainable Landfill Waste Forecasting

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Declaration

I declare that this Applied Research Project that I have submitted to Dublin Business School for the award of MSc in Artificial Intelligence is the result of my own investigations, except where otherwise stated, where it is clearly acknowledged by references. Furthermore, this work has not been submitted for any other degree.

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Note on this public version

This copy of the report is shared for portfolio purposes; personal details have been removed. Erratum: after submission it was found that the deployed web application loaded a stale LSTM checkpoint (best_model_lstm.pth) whenever that file was present, while labelling its output as Linear Regression. The California 2027 forecast quoted in Sections 5.6 and 6.1 (38,778,706 short tons) and the sample values shown in the appendix screenshots were therefore produced by the LSTM. The equivalent Linear Regression forecast for California 2027 is 36,113,367 short tons. The model comparison in Table 1 and all evaluation metrics are unaffected. The corrected application is available at github.com/alokekissac/AI-Landfill-Waste-Forecasting.

Abstract

The amount of landfill waste produced can differ considerably from one year to the next and from one state to the next in the United States, and it is important to forecast it accurately for sustainable landfill capacity planning. This work formulates a predictive modeling pipeline based on the Landfilled Waste Composition Dataset aggregated by state, year and with engineered lag (lag1, lag2, lag3) and rolling-mean features based on the historical waste mass. Five models were trained and compared: (1) Linear Regression, (2) Random Forest, (3) Gradient Boosting, (4) XGBoost, and (5) a two-layer LSTM network trained with PyTorch that uses dropout regularization and adaptive learning-rate scheduling. The held-out test set was used to assess the performance of the models using RMSE, MAE and R^2 . The results of Linear Regression were found to be the best ($R^2 = 0.986$, RMSE ≈ 1.05 M tons, MAE ≈ 0.58 M tons) compared to the ensemble and deep learning approaches, suggesting that the trends of waste at the state level are mostly linear, even with the inclusion of lag features. The trained model, encoders and scalers were then serialised and deployed in a flask webapplication providing interactive forecasting of waste up to 2035 for every state. The results show that lightweight interpretable models can be useful for landfill capacity forecasting, providing a viable low-cost decision support tool for sustainable waste management planning.

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Chapter 1: Introduction

1.1 Background

The United States' solid waste production has been steadily increasing over the past several decades as a result of the growing population, urbanization, industrialization, and consumption trends. This expansion puts a strain on landfill facilities the main means of final disposal in most states.

Landfill capacity is limited and the amount of waste sent to landfill fluctuates significantly, from state to state and from one year to the next. This fluctuation is attributed to the economic activities, seasonal fluctuations, industrial activities, population size and waste management policy in the region (Faiz, Ninduwezuor-Ehiobu, Adanma and Solomon, 2024).

Properly forecasting the quantity of waste for landfill is the key element to sustainable planning. The lack of reliable forecasts makes it hard for municipalities and state agencies to plan landfill expansion, to allocate the resources for collection and transport and to create long-term waste reduction strategies. Both mistakes in forecasting can be quite expensive in terms of dollars and the environment: one can be over-expanding landfill capacity while the other can be under-expanding and running out of capacity.

Artificial Intelligence (AI) has thus become a viable solution to these problems. In various waste management scenarios like municipal, industrial, and ocean waste, machine learning and deep learning algorithms are used for waste generation prediction, waste sorting, collection routing and waste disposal optimization (Rao, Singh, Salas, Sarker, Kumar, Wang, Lucia, Mittal, Yarbrough, Barlaz and Singh, 2025; Adeoba, Pandelani, Ngwangwa and Masebe, 2025).

In particular, predictive modelling has been found to be useful to help predict volumes of waste in the future so that landfill space can be allocated, resources can be planned and mitigation measures can be designed in advance (Ono and Okeagu, 2026; Adewuyi, Adebayo, Adebayo, Kalinzi, Ugiagbe, Ogunraku, Samson, Oladele and Adeniyi, 2024).

A wide overview of the use of AI in waste management has been provided in a number of studies, which have included algorithmic solutions, system architectures and large-scale implementations. In the case of time-dependent waste forecasting, regression-based models, ensemble tree models, and recurrent neural networks (RNN) are the commonly used approaches in these reviews (Divate and Chandrakar, 2026; Belyamani, 2025; Ahmed, Reddy and KM, 2025).

In environmental engineering applications, complex and non-linear relationships between environmental outputs and inputs have been modeled using machine learning algorithms like linear regression, random forest and gradient boosting (Kumar, Pandey, Saxena, Tiwari and Mishra, 2025; Talaie, Kamyab and Razmfarsa, 2024).

Where the underlying data exhibits some sort of sequential or time-series pattern, deep learning architectures, such as recurrent networks (e.g., Long Short-Term Memory (LSTM) models) have been applied. These models can be used to account for temporal dependencies, which simpler regression models might not fully represent (Lakhout, 2025).

For the same reason, these ensemble models like XGBoost have been used in environment and sustainability areas due to their capability to capture non-linear interactions between categorical and numeric features and the models are also not heavy on the memory, even at moderate dataset sizes (Gurjar, Kumar and Kuila, 2025).

The integrated waste management systems of smart cities and IoT have also shown how forecasting based on artificial intelligence can be integrated into the operational decision support tools, providing timely insights to the planners at municipal level and minimizing the manual estimation errors (Devi, Singh, Thandra, Gupta, Salman and Natarajan, 2025; Nabawagga, Birungi, Singh and Dash, 2026).

Big data analytics has also been useful in the identification of waste trends on a long term basis for policy-level planning, not only for short term operational decision making (Adewuyi, Adebayo, Adebayo, Kalinzi, Ugiagbe, Ogunraku, Samson, Oladele and Adeniyi, 2024).

In addition to forecasting, there is a variety of other sustainability applications related to artificial intelligence, such as optimization of vermicomposting, decentralized wastewater treatment and participatory environmental management. It signifies a larger trend towards data-driven solutions in the field of environmental engineering, not just for waste sorting, but also for other purposes, such as those listed in the introduction to this section (Goswami, Deka, Devi, Sarma and Brahma, 2024; Olukanni, Murhunzi and Mkandawire, 2027; Santos and Carvalho, 2025).

Regional case studies also indicate that the level of AI adoption in solid waste management depends on the maturity level of the infrastructure, the availability of data, and the capacity of institutions, emphasizing the importance of data-driven and context-specific forecasting studies, instead of generic frameworks (Gurjar, Kumar and Kuila, 2025).

Based on this growing number of works, the application and comparison of more than one predictive modeling technique specially at national (state) level for landfill waste forecasting is warranted, and should be backed by historical waste composition data, as is done in this work, rather than merely conceptual or review discussion.

State-level forecasting is especially significant because the administration of waste management policy in the United States is quite localized, and thus happens at the state level. The permitting of landfills, the approval of landfill capacity expansion and diversion targets are usually determined at a state level and not at a single national level (Faiz, Ninduwezuor-Ehiobu, Adanma and Solomon, 2024).

One way of forecasting which aggregates data at the national level and which does not maintain state-level detail, may obscure the key differences in waste trends at the regional level. The waste mass in a state could rise over time in relation to its growing population or expanding industrial activity, in other states it may remain stable or even decline over time and this may not be captured by the national model (Adewuyi, Adebayo, Adebayo, Kalinzi, Ugiagbe, Ogunraku, Samson, Oladele and Adeniyi, 2024).

In forecasting problems, features are frequently time-based, with lagged values and rolling averages being typical examples that provide information about recent movements and filter out the “noise” between years. However, these are effective only when the structure of the data is proper, and have not been systematically tested specifically for landfill waste, although there have been discussions about them in general environmental forecasting literature (Kumar, Pandey, Saxena, Tiwari and Mishra, 2025).

From a practical point of view, it is also important to make a comparative evaluation between the different families of models. The increase in complexity of such models as deep learning, which can include LSTM networks, often comes at a high cost of increased data requirements and longer training times, only compelling when there is a tangible improvement in forecasting accuracy (Lakhout, 2025; Divate and Chandrakar, 2026).

1.2 Problem Statement

While there is a growing traction on AI powered waste management solutions, many of the current studies are conceptual or review based that provide framework and bibliometric analysis, and not state level validated predictive models using historical data from landfills (Islam 2026; Mandal, Kundu, and Mondal 2024).

Data about the composition of landfill waste can easily be high dimensional, with categorical variables like state, type of facility, and type of waste stream as well as numeric variables like year and mass of waste. While some previous AI-based research on waste management has focused on this type of data, there is a lack of consistent treatment of the data in the form of aggregation and time features using lagged time steps (Goswami, Deka, Devi, Sarma and Brahma, 2024).

Raw waste data is often riddled with formula artifacts, missing data, non-positive mass data and inconsistencies in application of categorical labels. These problems, if not addressed in the preprocessing step, can impact the accuracy of the model training and result in inaccurate or misleading predictions (Ahmed, Reddy and KM, 2025).

Few works compare different types of models (e.g. linear models, ensemble tree based models, deep learning sequence models) on the same test set and hence with the same evaluation metrics. This is because it is not easy to identify a specific type of waste data with a particular model type especially at the state-year aggregation level, as adopted in this study (Gurjar, Kumar and Kuila, 2025).

There are a number of published studies that highlight the theoretical benefits of deep learning time-series forecasting models, like LSTM networks. The empirical performance of such models is however greatly influenced by the size of the datasets, the length of the sequences as well as the amount of noise, which are not necessarily tested in the waste management domain (Lakhout, 2025).

There is also limited work to translate trained forecasting models to be able to deploy and use. Some studies do not show an end-to-end system, but focus on the conceptual aspects of AI applications (Wadankar, 2026; Tamrakar, 2026).

Many predictive analytics frameworks described in the literature are conceptual or architectural in nature and do not indicate if an evaluation was conducted using real data from a public source, or what the result of the evaluation was, i.e., RMSE, MAE, R2, etc. (Ono and Okeagu, 2026).

In addition, the AI systems currently in the literature that are related to sustainability are mostly focused on the collection, sorting or recycling processes and there is very limited information on forecasting the waste mass in the final landfills, which is a critical input for capacity planning decisions (Reddy and Charhate, 2025; Belyamani, 2025).

This leaves a gap in translating AI research for waste management towards practical and usable forecasting tools for genuine decision making for sustainable landfill capacity planning. Such a study needs to involve a data preprocessing with soundness, multi-model comparison with transparency of evaluation and functional deployment in a single, reproducible pipeline.

Another drawback of the literature is that it is not clear what criteria are used to choose among models. A single model is chosen, and many studies rely on the use of the same model, possibly in advance, without providing justification for its selection compared to other models in terms of standard error metrics (RMSE and MAE) and goodness-of-fit metrics (R2) (Santos and Carvalho, 2025).

Another area that is not clearly defined is deployment. Very few systems documented in review papers and conceptual frameworks produce working software artifacts like saved model files,

encoders, and scalers for easy reuse and extension of a forecasting pipeline by future researchers/practitioners (Tamrakar, 2026; Wadankar, 2026).

If state agencies or municipal planners cannot access the model, even a successfully created model with a great deal of accuracy is not very useful for them, regardless of its accuracy. Therefore, a usable forecasting tool needs to provide an accurate forecast as well as an interface for non-technical stakeholders to request forecasts for a particular state and year (Belyamani, 2025).

1.3 Research Question

What is the maximum percentage of historical landfill waste composition data aggregated at the state level by the year that can be used to develop an accurate and deployable predictive model (chosen by RMSE, MAE and R2) for sustainable forecasting of landfill waste up to the year 2035 using a functional web-based forecasting tool by comparing the performance of Linear Regression, Random Forest, Gradient Boosting, XGBoost and two-layer LSTM network?

1.4 Objectives

The aim of this study is:

To clean, process and do aggregation on this Landfilled Waste Composition Dataset at the state-year level, which involves removing any formula artefacts, discarding any mass values that are not positive and cleaning categorical data, and engineering lag and rolling-mean attributes based on the history of waste mass.

2. To investigate and assess five predictive models (Linear Regression, Random Forest, Gradient Boosting, XGBoost, and two-layer LSTM with Dropout regularization and Adaptive learning-rate scheduling) based on RMSE, MAE and R2 on a hold-out test set.

3. To determine the best-performing model based on the comparative results of the evaluation and to analyze the reasons why a more simple model (linear) outperforms more complex models (ensemble and deep learning) on this specific dataset.

4. To serialise the selected model, as well as the various encoders and scalers used and to deploy the model using a flask based web application to generate interactive state-level waste forecasts up to 2035.

5. To assess the feasibility of the implemented forecasting model as a low-cost, sustainable landfill capacity planning decision support tool, and as part of the broader process of implementing AI research into operational waste management solutions.

Chapter 2: Literature Review

2.1 Introduction

This chapter covers relevant literature regarding the AI applications in waste management systems and focuses on predictive modelling, forecasting, machine learning and deep learning approaches, smart city applications and the implementation of AI tools.

The chapter has been divided into thematic subsections. Subsections are included to cluster studies together for comparison of methods, results and limitations. The chapter ends with a discussion of the research gap identified from these studies which will be the basis of the objectives presented in Chapter 1.

2.2 Overall assessments of the use of AI in waste management

2.2.1 Scope and Trends

However, there have been several studies that give a general overview of the use of AI in waste management. Belyamani (2025) explored the application, challenges, and opportunities of AI in waste management systems, detailing how AI is used across various aspects such as waste collection schedule optimization, automation of sorting, and forecasting waste disposal. These applications include optimizing waste collection schedules, automating sorting processes, and predicting waste disposal, while Belyamani (2025) discussed how, despite these advances, many AI applications in waste management are still in their pilot stages.

Ahmed, Reddy and KM (2025) carried out a literature review on sustainable waste management specifically, and found that machine learning approaches are gaining momentum over rule-based approaches due to the adaptability of machine learning algorithms to the change in composition patterns of waste over time.

Divate and Chandrakar (2026) conducted a systematic review of algorithms and integrated architectures for smart waste management, and classified them into supervised learning approaches, deep learning approaches, and hybrid architectures. They highlighted the fact that the

choice of algorithm might not always be the same depending on the data set used and noted that there is no algorithm that consistently outperforms other algorithms for all waste management tasks.

2.2.2 Bibliometric and Framework-Based Studies

Islam (2026) conducted a systematic review of the AI empowered framework and sensor-robotic network for urban waste management under the umbrella of smart city applications. The review revealed that the majority of the frameworks are conceptual, and have not yet been tested with actual waste data.

Adeoba, Pandelani, Ngwangwa and Masebe (2025) undertook a bibliometric analysis of the application of artificial intelligence in tracking and managing waste in the oceans with a view towards sustainability. Despite the different waste domain studied, the results of this study revealed that the same trends emerged as in the studies on terrestrial waste management, with increasing number of publications and lack of standardization of metric used for evaluation in the studies.

In the data-driven economy, Ono and Okeagu (2026) suggested the use of predictive analytics frameworks as a waste management strategy and not as a waste analysis tool. But their work doesn't show concrete performance of their models for a real data set.

2.2.3 General Review Common Themes

These views in the aggregate suggest three themes. The first theme is the rapid growth of AI in waste management, with varying rates of deployment in different areas and regions of waste management. The second theme is that the adaptive modeling is also constantly mentioned as a priority area, but in many cases this is not really empirical work, but rather a direction to move in. The third theme is that there is a wide spectrum on how to evaluate a study – some studies have a qualitative discussion, others report formal error measures and it is difficult to directly compare between literature.

These themes all point to the value of a study that would not only provide a multi-model comparison with rigor, but would also provide a working deployment that is more than just a descriptive study, like the general reviews available to date.

2.3 The Methods of Prediction and Forecasting

2.3.1 Landfill-Specific Predictive Modeling

Faiz, Ninduwezuor-Ehiobu, Adanma and Solomon (2024) directly discussed the topic on predictive modeling for sustainable landfill operations, using AI-driven forecasting techniques to predict landfill waste trends. Their work is very similar to the goals of this study, which focus on the need to predict the waste mass in landfills and not just sorting results and waste composition.

Their results confirm the premise that careful consideration of historical trends is needed to make landfill-specific forecasts, as waste generation trends may change in response to changing policy, economy and state-to-state migration.

The study also observed that long-term prediction (several years ahead) is advantageous to the landfill operation, because the planning and permitting process can often take a considerable amount of time. This helps to validate the design decision to produce a forecast of up to 2035, instead of restricting forecasts to one of the following year.

2.3.2 Big Data und Trend Forecasting

Adewuyi et al. (2024) used big data analytics to predict waste in the future and to guide waste planning for sustainability. They found that multi-year, state-level waste aggregation can be useful in forecasting pipelines since it can show long-term trends that can't be seen in short-term, localized data sets, confirming the value of aggregating large amounts of historical waste data.

Rao, Singh, Salas, Sarker, Kumar, Wang, Lucia, Mittal, Yarbrough, Barlaz and Singh (2025) gave a detailed review of application of AI in waste management of municipal solid waste in its entire life cycle from generation to utilization. The review showed that these forecasting models are more suitable when applied together with appropriate measures of evaluation like RMSE and MAE as well as qualitative assessment.

The review also identified the stages of the waste value chain as generation, collection, processing, and utilization, highlighting that a majority of AI applications are focused on the collection and sorting stages, while forecasting at the generation stage has been relatively less addressed, and was mainly done in a qualitative and rigorous way. This imbalance makes it important to devote a study only to prediction of waste mass at the landfill disposal site.

2.3.3 Optimization of Waste Management with the Help of AI

Reddy and Charhate (2025) discussed the potential benefits of AI in optimizing sustainability outcomes in the waste management industry, particularly in the context of predictive analytics for enhancing operational efficiency. They found their study re-emphasised the need to transform forecasting data into actionable planning decisions and not just a technical exercise.

Kumar et al. (2025) discussed the sustainable solutions using AI in environmental engineering in the areas of monitoring, modeling, and mitigation. Their work put predictive models in perspective, as part of a wider environmental management system, not only monitoring, but also a mitigation system.

In sum, the studies reviewed in this section point to the fact that the use of predictive models has always been acknowledged as key to the sustainable management of waste, whatever it is presented as: Landfill forecasting, trend analysis or operational optimization. But the way state level aggregation is done, the use of lagged variables, and the comparison of multiple models as applied

in this study does not specifically exist in any single reviewed study, so it is deemed as a meaningful methodological contribution.

2.4 Machine Learning and Deep Learning Techniques

2.4.1 Traditional and Ensemble Machine Learning Models

Talaie, Kamyab and Razmfarsa (2024) examined the contribution of AI to development in environmental engineering, which is a field where traditional machine learning models, such as regression, still play a significant role in usage due to their ease of interpretation and less computational demands than alternatives such as deep learning.

Gurjar Kumar and Kuila (2025) have done a study on the present scenario and future scope of Artificial Intelligence in solid waste management in India. Ensemble models like random forest and gradient boosting have been widely used, as waste composition data often includes a combination of categorical and numerical variables, the review stated.

Another key finding of the review is that gradient boosting frameworks, such as XGBoost, are often preferred for applied work in the environmental field due to the fact that they typically offer competitive accuracy, while requiring fewer hyperparameters to tune than deep learning. This observation is pertinent to the comparative model evaluation performed in this study in which boosting-based methods are compared with simpler and more complex ones.

2.4.2 Deep Learning and Sequential Modeling

Lakhouit (2025) surveyed the role of AI and IoT in smart waste management systems, focusing on smart waste collection, sorting, and recycling, and found a growing number of researchers interested in investigating deep learning models, such as recurrent, for sequential and time-related waste data. The review warned of the need to have large and consistent historical datasets to make the deep learning models superior to a simpler statistical or machine learning method.

This is especially important, for example for time-series forecasting (e.g., with Long Short-Term Memory networks), where the number of available sequences per category (here: number of years available for a state) is small, and therefore the number of sequences is not sufficient to make the recurrent models the preferred choice over linear or tree-based models.

This is a problem especially in the presence of short time series (few decades of data) for each of the states, as recurrent networks like LSTM networks typically need longer time series to take full advantage of their ability to capture long-range temporal dependencies. If the length of the sequence is short, simpler models, which need very few lag features, can prove to be more generalizing than deep sequential models.

2.4.3 AI-Driven Environmental Management

Santos and Carvalho (2025) delved into AI-powered participatory management of the environment, examining innovations, applications, and future perspectives in environmental areas.

In their study, they stressed how transparency and interpretability of the model are crucial when the AI system is designed for supporting decision making at the public sector or at the policy level, which makes the comparison of the interpretable model (linear regression) to the other more complex models relevant.

This focus on interpretability is well in line with the findings in the literature reviewed of the machine learning field, and the literature shows that in public sector or policy facing applications simpler, more transparent models are often favoured, despite the fact that only a very small improvement in accuracy is gained with a more complex model. This consideration directly impacts the analysis process used in this research – the analysis method is based on the practical interpretation of the model results, and the statistical interpretation of the model results as well.

2.5 Smart City and IoT-Integrated Waste Management Systems

2.5.1 Sensor and IoT Based Systems

Devi, Singh, Thandra, Gupta, Salman and Natarajan (2025) proposed a smart urban waste management system that utilizes AI and IoT technologies for efficient waste collection, segregation, and disposal. While the study showed that sensor data combined with predictive modelling can help to increase the responsiveness of the operation, it was mainly aimed at collection logistics and not on long term forecasting of landfill capacity.

Although sensor data was not directly available in the scope of the present study, the study illustrated that IoT sensor networks generate large volumes of data to be transmitted as near-real-time information which could be used to complement the historical, aggregated data to provide finer-grained input for short term operational decisions.

The study by Nabawagga, Birungi, Singh and Dash (2026) explored the potential use of AI technologies for smart waste management in smart cities with a focus on integration with economic factors and energy recovery. They pointed out that, because of their work, forecasting was no longer a task to be carried out as an analytical exercise but rather one component of an integrated set of components.

They discussed how their forecasting outputs are not necessarily a standalone output, but are often incorporated into other municipal budgeting and resource allocation processes as well, along with an economic stability component. This underscores the importance of a deployed tool that can generate forecasts that can be easily queried by decision makers for particular states and years.

2.5.2 Deployed AI-Powered Systems

Tamrakar (2026) presented the design of AI-based waste management system, which detailed the architecture of the system and its functional elements but not in depth comparison of these models. Likewise, Wadankar (2026) introduced a smart waste management system that involves artificial intelligence (AI) and focuses on the practical implementation of the system, such as user interaction and system usability.

Both studies show that functional deployment – accessible interface for end users – is an important but underdeveloped part of AI-powered waste management research, which further strengthens the need to use an accessible interface for forecasting in this study.

Details of underlying predictive algorithms were not reported or compared, however, nor were standard error measures reported for the models contained within the systems of either of these studies. This indicates that the algorithmic component that produces good forecasts has been much less documented, and thus there is still a study that could be conducted that treats both model comparison and deployment in the same manner.

2.6 Sector-Specific and Related Applications

2.6.1 Energy Conservation and Management System

Goswami et al (2024) have discussed the AI assisted vermicomposting systems for organic waste management and income generation, from the perspective of Guwahati, Assam. While the study was not specifically a forecast for landfill but a study on how predictive and optimization techniques can be adapted in different subdomains of waste management, it showed the applicability of prediction and optimization techniques in waste management.

The study also showed that AI driven optimization can contribute to income generation as well as the environment, indicating that predictive tools in waste management have an economic, as well as environmental value – which can also be relevant for the justification of investing in predictive waste management infrastructure at a state level.

2.6.2 Applications for Water and Wastewater

In decentralized wastewater treatment systems, Olukanni, Murhunzi and Mkandawire (2027) have explored innovations enabled by AI for optimizing the treatment process and promoting sustainable water management. They conclude that methods used for predicting mass in one environmental area (water treatment) are similar in other areas such as waste mass forecasting, such as using historical time-series data and regression-based modeling.

The similarities between water treatment optimization and waste mass forecasting also add to the transferability of the modeling techniques assessed in this study; models and modeling techniques found to be effective in one forecasting problem are often transferable to other environmental time-series forecasting problems.

2.6.3 General Waste Management Innovations

Mandal, Kundu and Mondal (2024) did a review on the innovations in the field of waste management which includes technological, regulatory and operational changes. Their review gave background information to comprehend the role AI-based forecasting plays in a more broader array of innovations in the waste management sector, such as recycling technology and policy reform.

The review findings highlighted that innovation in waste management cannot be achieved through technical solutions only and are reliant on institutional take-up, data availability, and policy alignment. This observation is applicable to the current study as a basic precondition for creating reliable forecasts is the existence of a structured historical state level data set, which is not used merely as a side product.

2.7 Comparative Summary of Reviewed Studies

2.7.1 Methodological Diversity

The studies examined in this chapter have a variety of methodological strategies, ranging from bibliometric description to applied predictive modeling. To circumvent this, bibliometric and systematic reviewing studies like Adeoba, Pandelani, Ngwangwa and Masebe (2025) and Divate and Chandrakar (2026) are conducted which involves analysing citation and publication pattern and not training the model itself. Other studies like Faiz, Ninduwezuor-Ehiobu, Adanma and Solomon (2024) directly use predictive modelling techniques to waste data and were more direct in comparison with the approach adopted in this study.

The methodological variety makes it hard to make general conclusions for the literature as a whole. A bibliometric study, for instance, can show that AI-based forecasting is an emerging research field, but not which is the most suitable algorithm to use on a particular type of waste data. The usage of the models is hence more directly related to the model selection decisions in the research.

2.7.2 Dataset Scope and Granularity

There was also a wide range of data sets that were studied in the literature reviewed. Some studies like Rao, Singh, Salas, Sarker, Kumar, Wang, Lucia and Mittal, Yarbrough, Barlaz and Singh (2025) have considered the municipal solid waste at a wider scale – aggregate level, and others, like Gurjar and Kumar, Kuila (2025) have considered municipal solid waste at a narrower scale – regional/country specific. In the past, there are very few studies which address specifically landfill waste mass at state-year granularity that is applied in this study.

This granularity gap is important as the precision of the forecast and the appropriate level of complexity of the model can vary significantly between different levels of aggregation – national, regional or at a more detailed state level. The aggregation may affect the meaningful regional variation, as coarser aggregation can smooth out the regional variation while finer aggregation can lead to more noises in the data, which then need to be solved by feature engineering, such as the lag and rolling-mean features in this study.

2.7.3 Evaluation Consistency

Different practices of evaluation are also followed. Some of the reviewed studies, for instance Santos and Carvalho (2025) and Kumar et al. (Pandey, Saxena, Tiwari and Mishra 2025), have largely discussed the performance of AI in qualitative terms, and others, for instance Rao et al. (2025), have mentioned quantitative measures without providing a full table of multi-model

comparisons. This lack of uniformity in studies makes it difficult to directly compare the current study with one other study; it is therefore necessary to reference various studies as a whole to support the methodological decisions made in this chapter, as illustrated.

2.8 Research Gap

Throughout the literature that was reviewed, there are a number of common areas that directly inspire this study.

Firstly, most of the studies discussed are conceptual, review-based, and framework-oriented studies, which lacked empirical results in training and testing the model using a specific data set (Islam, 2026; Mandal, Kundu and Mondal, 2024; Ono and Okeagu, 2026).

Secondly, the cross evaluation of different model families is still limited. Most studies either study one type of model or talk about multiple approaches without the direct comparative evaluation of the models using the same performance metrics (such as RMSE, MAE, and R2) on the same data set (Gurjar, Kumar and Kuila, 2025; Divate and Chandrakar, 2026).

Thirdly, there are few studies on landfill-specific forecasting at the state/regional level in comparison to the more general municipal solid waste management studies, although it is directly relevant to landfill capacity planning (Faiz, Ninduwezuor-Ehiobu, Adanma and Solomon, 2024; Adewuyi, Adebayo, Adebayo, Kalinzi, Ugiagbe, Ogunraku, Samson, Oladele and Adeniyi, 2024).

Fourth, deep learning methods like LSTM networks are often suggested but not compared to simpler statistical methods to assess the need for increased complexity for a data set of a particular size and structure (Lakhout 2025; Talaie, Kamyab and Razmfarsa 2024).

Fifthly, the use of trained forecasting models is still limited. There are multiple research papers that discuss the system design or conceptual design without showing a full end-to-end system for the processing of the raw datasets and the provision of an accessible forecasting interface (Tamrakar, 2026; Wadankar, 2026; Belyamani, 2025).

Sixth, the measurement of the effectiveness of assessments in the literature studied is also inconsistently reported. Only a few studies mention RMSE, MAE or R2 in the general context; few compare the model performances objectively in terms of numbers by presenting side-by-side comparison table of various candidate models trained on the same training data (Rao, Singh, Salas, Sarker, Kumar, Wang, Lucia, Mittal, Yarbrough, Barlaz and Singh, 2025; Santos and Carvalho, 2025).

Seventh, conceptually in other environmental fields like wastewater treatment, lagged historical values and rolling averages are mentioned when dealing with waste time-series data, but the same are not necessarily documented explicitly for landfill waste mass forecasting (Olukanni, Murhunzi and Mkandawire, 2027; Kumar, Pandey, Saxena, Tiwari and Mishra, 2025).

2.9 Summary

The literature review has revealed that the use of AI in waste management is well acknowledged and has been used in various applications ranging from waste collection, to landfill capacity planning. But there is still a small number of studies which specifically focus on landfill waste mass forecasting at State level from empirical, comparative and deployment perspective.

Building on the themes and limitations of this chapter, this study aims to: 1) address the gaps in the structured data preprocessing, 2) apply a lag-based feature engineering method, 3) compare different models using five distinct algorithms, 4) provide a transparent evaluation with standard regression metrics, and 5) deploy the model in a web-based forecasting app.

The literature covered in this study includes general AI reviews, landfill and municipal forecasting studies, comparisons between machine learning and deep learning methods, smart city and IoT integrated systems, and applications to specific sectors, such as, organic waste management and wastewater management. In all these areas, there is a consistent description of the usefulness of predictive modeling, but it is not empirically and comparatively evaluated with that of landfill waste forecasting.

The methodological decisions made in this study, such as the experience of state-year aggregation, the features engineered as lag/rolling-mean, and the use of five families of models, and the deployment in an interactive web application, all explicitly address the gaps identified in this chapter. The methodology for the implementation of these choices in detail is presented in the following chapter, including the data used, model architecture, training strategy, and evaluation strategy.

Chapter 3: Methodology

3.1 Introduction

The methodology to develop the landfill waste forecasting system is presented in this chapter. It includes research design, description of the data set, the initial data processing steps, feature engineering, the development of the model(s), training procedures, evaluation metrics, and how the model(s) were selected and persisted. All the steps explained in this chapter are the same as those shown in the data processing notebook and deployment script for Flask.

3.2 Research Design

Research design used is quantitative, experimental design. A number of predictive models are trained on historical data on a landfill waste and the performance of each model is evaluated using the standard regression measures. The design is of the comparative type, that is, five different models are trained on the same dataset after processing and tested in the same train-test conditions, so that they can be compared. This comparative structure enables the comparative evaluation of the three different deep learning approaches: linear, ensemble and sequential.

There are 5 stages to the overall workflow. There are two stages—data acquisition and data cleaning. The second phase is the feature engineering and aggregation. The third stage is training of the model that is done with the five algorithms. The fourth stage is the evaluation and the selection of the model. How the chosen model will be deployed via a web application is the fifth stage. The system architecture is shown in figure 1, with the following stages of raw datasets ingestion to deployed forecasting interface.

3.3 Dataset Description

This study was based on the Landfilled Waste Composition Dataset which was provided in an excel sheet. The raw data includes such information as State, Year, Waste Mass (short tons), ORD-

Designated Facility Type, ORD-Designated Waste Stream and Calendar or Fiscal indicators, to mention a few.

The data is loaded from the database using `openpyxl` with the option `openpyxl.data_only=True` which means that the cached formula result from the workbook is read, rather than the formula itself. This is important as some of the columns in the original excel file have `XLOOKUP` formulas and not set values. The column headers are trimmed of any leading/trailing white spaces and columns with an invalid column header name are not included in the working dataframe for further processing.

3.4 Data Preprocessing

The first step in data preprocessing is to remove the rows with formula artefacts. In particular, formula strings (those beginning with an equals sign) are ignored in the ORD-Designated Facility Type and ORD-Designated Waste Stream columns because they are not categorized.

Then, the Waste Mass (short tons) and Year columns are coerced to numeric, using pandas's `to_numeric` with error coercion which will turn any bad data into a missing value. Any row that has no waste mass, year or state is removed from the data.

Note that only records that have a positive value for mass of waste are kept as this will not be a valid disposal event for a record with zero or a negative value for mass of waste. After this cleaning step the type of the Year column is cast to integer.

Extreme outliers of the target variable are dealt with by capping values at the 99.9th percentile so that a few very high waste mass values would not have a huge effect on the training of the model, but would not completely remove the records. The skewness of target is calculated before and after this capping process, which assures that the resulting distribution can be used for regression modelling better than the raw values without the capping process.

3.5 Feature Engineering

The data for Categorical variables (such as State, ORD-Designated Facility Type, ORD-Designated Waste Stream, and Calendar or Fiscal) are label-encoded so that they can be used with machine learning models. The encoders that are fit are saved for future use in the process of inference.

After encoding, the data are summarized according to State, State encoding and Year with the Waste Mass (short tons) adding up within each summary. This creates a time series of state-year level data for total landfill waste mass for each State for the years available.

The aggregated waste mass values are log transformed and creating a `log_total` feature. This change will help to remove skewness from the target distribution and help to equalise the variance between waste volumes across states that have widely varied waste volumes.

Lag features are created from the log transformed series, such as lag1, lag2 and lag3 which are the log of the waste mass from one, two and three years ago for each state. These lag values are also used to calculate a 3-year running mean (roll3). Another feature, yrs_since, is the number of years since the first recorded year for each state and represents long-term trend position of the training data. This is also the case for states with shorter historical time series; if there are fewer than three years of data for a particular state at the time of the prediction, then the earliest available lag is filled in for the missing lags, so that the feature vector is complete even for shorter time series states.

For the non-sequential models, the feature set is State encoding, Year, yrs_since, lag1, lag2, lag3 and roll3 and the prediction target is log_total.

3.6 Model Development

In this study, five predictive models are created and compared.

3.6.1 Linear Regression

Linear Regression model is trained as a basic linear approach. Before training, features are normalized by a scaler as well as target is scaled before fitting.

3.6.2 Random Forest Regressor

This study used Random Forest Regressor with 200 estimators, max depth 20 and minimum number of samples per leaf 5. This ensemble method learns non-linear relations among features as a combination of a number of decision trees.

3.6.3 Gradient Boosting Regressor

The scikit-learn Gradient Boosting Regressor has 200 estimators, a maximum depth of 6, a learning rate of 0.05 and a subsample ratio of 0.8. This is a second comparison model for boosting based on this model, along with XGBoost.

3.6.4 XGBoost Regressor

The model trained is an XGBoost Regressor with histogram-based boosting with 300 estimators. The training device will be put to use GPU in the event that it is offered, otherwise CPU.

3.6.5 LSTM Forecaster

PyTorch is used to build a two layer LSTM network. The architecture is comprised of an LSTM layer of hidden size 128 with two stacked layers, and then a dense head made up of a linear layer, ReLU activation, dropout regularization and a linear output layer. A three-window sliding window is applied to the state-year aggregated waste mass series to build up input sequences, and two different scalers are applied to input sequences and target values.

3.7 Model Training Procedure

In the non-sequential models, the feature matrix is split to train and test sets, and the train data is used for scaling the features using fitted scalers. Scaled training data is used to train each algorithm

– Linear Regression, Random Forest, Gradient Boosting and XGBoost – and then compared to the held-out test data.

For training, the LSTM model is trained with the Adam optimizer with an initial learning rate of 0.001 and 1e-5 for weight decay (regularization). ReduceLROnPlateau is a learning rate scheduler that reduces the learning rate if the validation performance doesn't improve. Gradient clipping is used to avoid exploding gradients in the back-propagation through the recurrent layers. The best model in the validation performance is checkpointed and loaded before the final model performance is reported, so that the reported LSTM results are not the results of the last epoch, but the results of the best epoch.

3.8 Evaluation Metrics

RMSE, MAE and R2 are three metrics used to evaluate the performance of the model on the held-out test set. RMSE and MAE are presented in the original waste mass scale, after inverse-transforming predictions if they were done in a scaled or log space, because the predictions of the Linear Regression and LSTM are done in a scaled or log space before transforming them back to raw tons.

These metrics are computed in all models, using a common evaluation function, which makes results reported for each model directly comparable.

3.9 Model Selection and Persistence

All five models are trained, tested and then all the results are summarised in a single table of metrics which is sorted according to R2 values from highest to lowest. The model with the highest R2 score is considered as the best performing model to be deployed.

The label encoders, feature scalers, target scalers, metrics summary, all trained models (sklearn based) and the LSTM model checkpoint are saved in a dedicated saved_models directory. The LSTM checkpoint also contains its architecture configuration, model state dictionary, related scalers and sequence length so that the model can be reconstructed exactly the same during deployment without the need for accessing the LSTM's training environment.

3.10 Deployment Methodology

A Flask web application is started and loaded with the selected model and model related artifacts. The state-year data is pre-aggregated and stored in a cached, pre-aggregated version as a CSV file in order to prevent reprocessing the entire Excel file with each start of the application.

During runtime the application re-creates the desired lag and rolling-mean features for the requested state and year from the most recent available historical data for the state, applies the desired scaler transformations, and makes a prediction using the requested model. Predictions are then published via a REST API link and accessed via a Web browser interface—from raw historical data to interactive forecasting.

3.11 Summary

The entire methodology for developing the landfill waste forecasting system has been described in this chapter, ranging from the preparation of the datasets to the feature engineering, training of the five different predictive systems, testing of the selected system using the same regression metrics and persistence of the selected system for deployment. The design specification and implementation details of deployed system are presented in the following chapter along with the system architecture as shown in figure 1.

Chapter 4: Design Specification and Implementation

4.1 Introduction

This chapter will explain the detail information of the design specification and implementation of the landfill waste forecasting system. It covers the general system design, the individual parts of the pipeline, the data transfers among the individual components and the implementation of the web-based deployment level.

4.2 System Architecture

The system architecture is depicted in Figure 1, which represents four main layers namely data layer, model layer, application layer and presentation layer.

Figure 1: System Architecture of the AI-Powered Landfill Waste Forecasting System

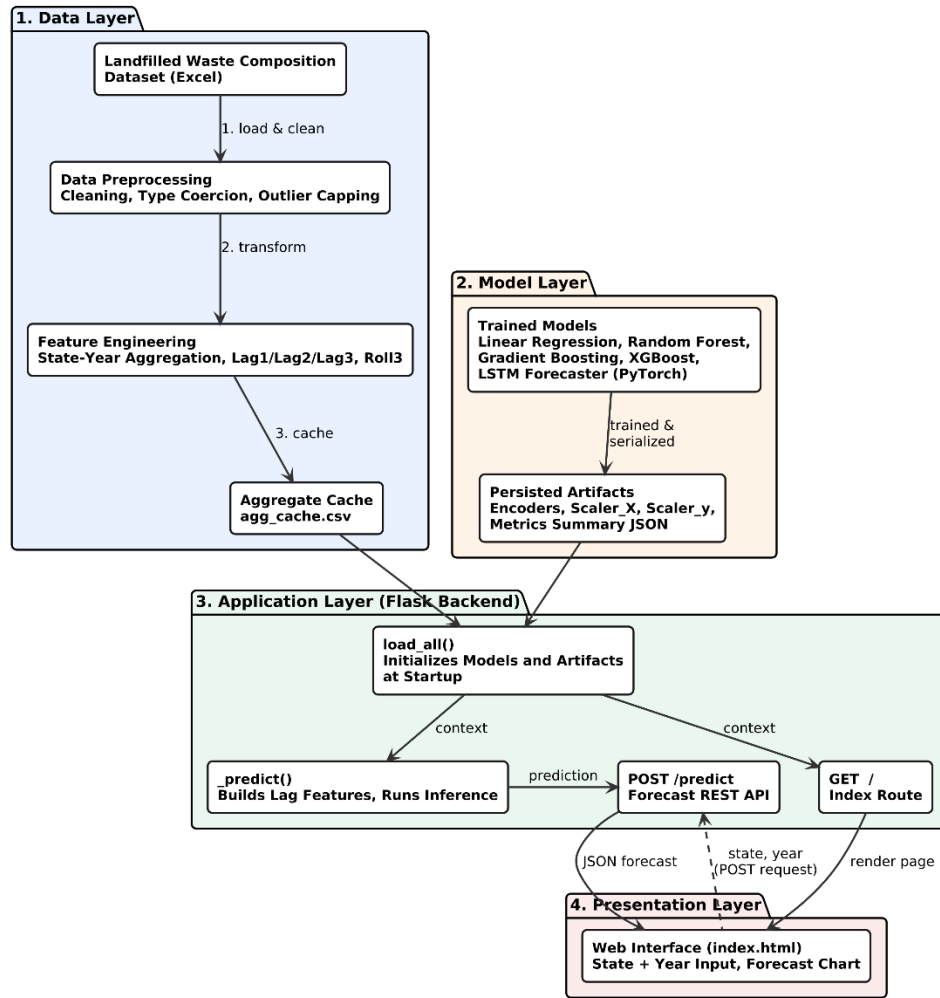


Figure 1 System Architecture

The data layer consists of the raw data from the Excel file, the logic to preprocess the data (cleaning, aggregation of data by year and state), and a cached version of the preprocessed state-year aggregates as a CSV file for rapid re-use. The five trained models, their corresponding encoders and scalers are saved as serialized files in the model layer. The application layer consists of a Flask backend which loads these artifacts, rebuilds these features upon receiving a request and produces predictions. The presentation layer is an HTML interface where a user can select a state and year, and see a forecast for that state and year and historical trends.

As seen in Figure 1, these layers are linked together in a sequential manner from the ingestion of the datasets to the preprocessing of the data, loading the model, generating predictions and displaying them in the web interface.

4.3 Component Design

4.3.1 Data Aggregation Component

A state-year aggregated version of the data is created at training and inference time with the `build_df_agg` function. First time it looks for the presence of a cached aggregate file. If the cache is available, the cache is loaded directly, thus preventing the processing of the entire Excel data set repeatedly. Otherwise, the function cleanses and aggregates all the data, transforms the State column using the label encoder that was already fitted, transforms the target column using the logs, and saves the result in the cache file (for future use).

4.3.2 Model Loading Component

The function `load_all` is the main initialisation function for the application. Loads the Label Encoders, Feature Scaler, Target Scaler, and Metrics Summary from the `saved_models` directory. It then looks for the existence of a checkpoint file created by LSTM. When an `LSTMForecaster` is found, its architecture parameters are loaded from the checkpoint and the weights in the model are loaded from the saved model. If there is no LSTM checkpoint, a serialized scikit-learn model is loaded and the type of the model is recorded.

4.3.3 LSTM Model Definition

The `LSTMForecaster` class specifies a two-layer LSTM network with a settable size of the hidden layers and dropout rate. The forward method is used to pass an input sequence through the LSTM layers and then pass the output at the last time-step through the dense head (linear layer, ReLU activation, dropout, and a linear projection) to get a predicted value.

4.3.4 Prediction Component

The `_predict` function is the base code used for predicting both at start up when generating an initial sample and in the live prediction API. In the case of sequential models, it loads the latest history data of the states requested and repeats the edges to pad the sequence if they are too short, it scales the input sequence and it feeds the sequence to the LSTM model in evaluation mode to get the prediction and converts it back to the original waste mass scale.

For non-sequential models, the function reconstructs the 4 features (lag1, lag2, lag3, roll3) from the latest records provided before the requested year, scales the feature vector, and provides a prediction, based on the earliest recorded year in the state. Special handling is performed for Linear Regression: The prediction is inverse transformed using the target scaler, and then converted back to the raw tons by an inverse log scaling.

4.4 Data Flow

The data flow starts with the Flask application start up sequence where `load_all` is called to load all the model artifacts and the aggregated data set. In this process, sample forecasts are calculated for the six states that historically have the greatest mass of total waste prior to the creation of the

homepage so that the homepage can provide representative forecasts without asking for a forecast for each individual state.

If the user makes a prediction request via the interface, the state and year chosen will be sent to the /predict endpoint as a JSON payload. Before invoking the prediction logic, the backend checks the requested state is in the known list of states and if the requested year is in a year range accepted by the backend. The forecast and whole historical series for selected state is returned as a json result and displayed in the interface.

4.5 API Design

The Flask application has two routes implemented. The index route is the route that renders the main interface, which includes the list of available states, the pre-computed sample predictions and the table of the metrics for the model comparisons. The /predict route is a POST method with a state and year input, does some validation and returns the JSON object of predicted waste mass, the model used, and the set of data points available for the state.

The /predict route also provides the correct JSON error messages with the proper HTTP status codes if the year entered is not in the right format, if the state is not recognized, or if the year entered is outside of the acceptable range from 1980 to 2035.

4.6 Implementation Environment

The backend is in the python programming language and was built using the flask web framework and PyTorch for the LSTM model while scikit-learn was used for the other four models. For scikit-learn artifacts, the persistence is based on joblib, while the LSTM checkpoint is natively saved by PyTorch. The frontend consists of a HTML template rendered on the server and which talks to the backend via asynchronous calls to the prediction endpoint.

4.7 Summary

This chapter has described the design specification and implementation of the deployed forecasting system, the layered system architecture as shown in Figure 1, individual components to aggregate the data, load the models, and generate predictions, and the API design that connects the backend with the user interface. These elements create an entire pipeline of raw waste data to a state-level interactive forecasting tool.

Chapter 5: Results

5.1 Introduction

The results of the landfill waste forecasting pipeline detailed in Chapter 3 is presented in this chapter. The results are presented for the processed data, the five trained prediction models, the comparison between the models based on RMSE, MAE and R2, and the forecasting application deployed behavior.

5.2 Dataset Composition After Preprocessing

The state-year dataset has been cleaned, aggregated and feature engineered to 846 records across 45 states and includes historical data from 1938 to 2023. Minutes and maximums of years of

records vary widely from state to state, with the minimum of 1 year of records for the least represented state, and the maximum of 84 years of records for the most represented state, for an average of about 18.8 years of records per state.

The mass of landfill waste collected for the total processed data set equals about 7.77 billion short tons, for all states and all years. California had the largest accumulated mass of waste disposed of at landfills, with an estimated 1.22 billion short tons of waste, followed by Texas with an estimated 1.04 billion short tons of waste, Pennsylvania with an estimated 723 million short tons of waste, Florida with an estimated 592 million short tons of waste, Michigan with an estimated 436 million short tons of waste, and Ohio with an estimated 397 million short tons of waste. These six states constitute the highest volume group of states for which samples are being generated for forecasts in the deployed app.

States with lower totals at the bottom of the list, including Connecticut, Guam, Nebraska and Rhode Island, have less historical data or perhaps, true lower short ton volumes of landfill waste disposed of. This is a huge difference in scale between states (over three orders of magnitude), and directly inspired the idea of applying a log transformation to the target variable, prior to training the model, which prevents high-volume states from dominating the loss function during training.

5.3 Comparative Model Performance

The performance of each of the five trained models on the held-out test set are summarized in Table 1 based on the R2 score.

Linear Regression had the best overall results with an R2 of 0.9861, an RMSE of around 1.05 million short tons and an MAE of around 0.58 million short tons as can be seen in Table 1. This is about 22 percent less RMSE and 24 percent less MAE than the second place Random Forest model.

Random Forest and Gradient Boosting were almost identical with a difference of only 0.0002 in R2 and a difference of ~5,073 short tons in RMSE, suggesting that both ensemble tree-based methods capture a similar variance in the target, despite the differences in ensemble construction and boosting strategy.

Although XGBoost is a gradient boosting family and has a GPU accelerated histogram model training, it generated a lower R2 of 0.9750, and higher error values when compared to Random Forest and Gradient Boosting. This finding suggests that the extra complexity of the optimizations XGBoost is able to perform does not translate to a boost in performance for this specific structure of data: a relatively small number of engineered lag features per state-year record.

The LSTM model had the lowest R2 score of all five models, 0.9547, and the highest RMSE value of ~1.90 million short tons and MAE value of ~1.10 million short tons. The LSTM error metrics are approximately 81% and 92% greater than Linear Regression in RMSE and MAE, respectively. The bar chart comparison of the R2 scores of all the five models is displayed in Figure 2, where the model with the highest score is clearly displayed, thus visually confirming the ranking shown in Table 1.

Table 1 Comparative performance of predictive models on the held-out test set

Rank	Model	R ²	RMSE (short tons)	MAE (short tons)
1	Linear Regression	0.9861	1,052,806	575,628
2	Random Forest	0.9792	1,287,503	703,022
3	Gradient Boosting	0.9791	1,292,576	754,781
4	XGBoost	0.9750	1,413,426	785,371
5	LSTM (PyTorch)	0.9547	1,901,529	1,103,571



Figure 2 R² comparison across the five trained models.

5.4 Error Metric Analysis

Figure 3 shows a comparison of the RMSE values of the models using a bar chart with the lowest error model again being the Linear Regression model. Figure 4 shows a side by side subplot displaying both RMSE and MAE for all models together for direct visual comparison of the extent to which the error profile of each model changes between the two error metrics.

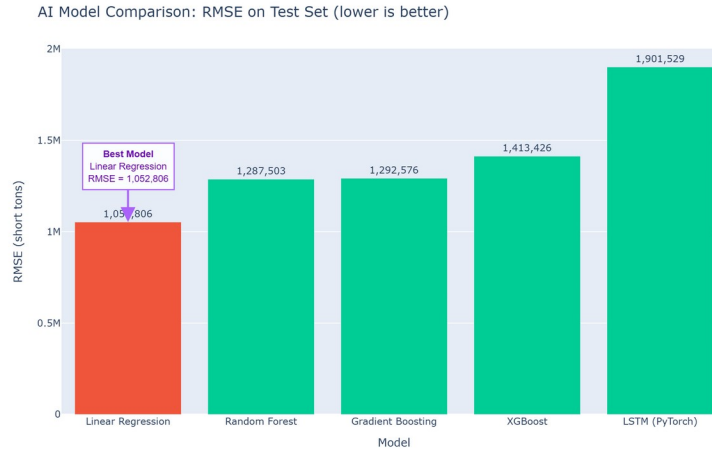


Figure 3 RMSE comparison across the five trained models.



Figure 4 RMSE and MAE comparison across the five trained models.

As can be seen in all the five models, the MAE values are consistently lower than the RMSE values because the RMSE values account for larger errors in predictions because of the squaring process in the formula. For the Linear Regression model, the difference between RMSE and MAE is smallest, with a value of about 477,178 short tons; and largest for LSTM, with a value of about 797,958 short tons. This larger range for LSTM indicates that the errors of LSTM predictions are not evenly distributed, but instead a portion of LSTM predictions has a significant deviation from the real value compared to the majority of LSTM predictions.

The stability of the RMSE-MAE gap for Linear Regression indicates that the errors are fairly well distributed over the test set, rather than a few extreme outlier predictions driving the error. This is in line with the previous decision to clip outlier target values to the 99.9th percentile when pre-processing, which decreased the effect of outlier waste mass values on training and evaluation.

5.5 Factor Analysis of the Model

An interesting finding from this work is that the performance of Linear Regression was better than four more complicated modeling methods: tree-based ensembles and a recurrent deep learning architecture. This is due to several factors relating to the structure of the processed data set.

First, this feature set, consisting of only state encoding, year, years-since-start and three lag values plus a rolling mean, is low dimensional as compared to the complexity that models based on trees and deep learning would typically be designed to exploit. The log transformed waste mass has limited non-linear structure after the log transformation – the relationship between the lagged historical variables and the waste mass of the current year seems to be mostly linear and limited additional non-linear structure is left to the ensemble or deep learning methods to capture.

Second, the average number of years of data (around 18.8 years) for each state is not large enough to enable the LSTM model to learn strong long-range temporal patterns, as some states have less than 10 years of data. Recurrent architectures generally need more and longer sequences to be superior to simpler statistical baselines and in this study, there was a relatively short and uneven historical coverage between states which has restricted the LSTM model.

Third, although more flexible than Linear Regression, Random Forest and Gradient Boosting might have overfitted a few state-specific patterns that were not consistently represented in the test split and consequently introduced a small amount of error, even though these models have a greater representational capacity.

Fourth, the input features and the target, which are in log-space, seem to have been well standardized before fitting the Linear Regression. After the waste mass values are compressed by the logarithmic transform described in Chapter 3, the relationship between the lagged values and the current period value is approximately linear and a linear model is able to capture the dominant signal without the need of any extra non-linear terms.

Fifth, the use of a relatively small training set size with the state-year data, when combined with iterative error correction using sequential tree fitting as performed by both algorithms, XGBoost and Gradient Boosting, can trigger overfitting. Both models quickly reached their peaks of performance with the smaller number of training examples for each boost, and were unable to beat the more parsimonious LR baseline due to this fact.

This overall ranking trend - a simple, well regularized linear model, compared to more complex ensemble and deep learning models, on a moderately-sized, feature engineered environmental dataset - is consistent with the observations made in Chapter 2 that more complex models would need larger and more consistent datasets to achieve their full representational benefit.

5.6 Sample State Forecasts

At application startup, it pre-computes the forecasts for the top six states that were identified in Section 5.2 based on volume. When using the Linear Regression model (which was determined to

be the best model in Table 1), a request was sent for a verification to the deployed prediction endpoint for California using a target year of 2027; the waste mass forecast returned was 38,778,706 short tons. This forecast was accompanied by a full historical series of data for California from 1990 to 2022, which ranged from a minimum of about 29.4 million short tons in 2012 to a maximum of about 59.1 million short tons in 1995.

Figure 5 shows the overall multi-decade trend line for total landfill waste mass across all states, by year, before disaggregating by state. The distribution of the top waste generating states according to the mass of historical landfill wastes is shown in Figure 6, based on the ranking mentioned in Section 5.2.

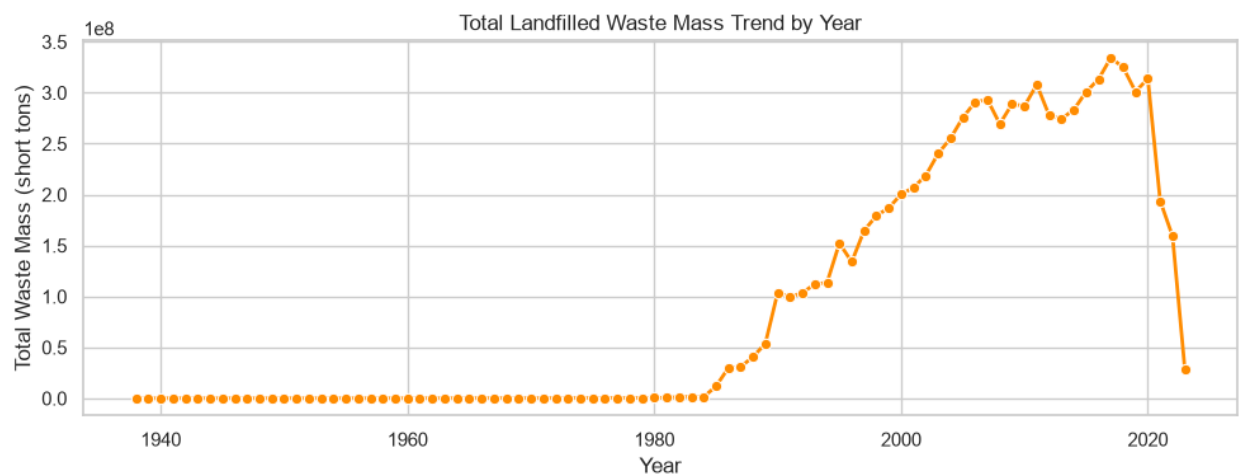


Figure 5 Total landfilled waste mass trend by year.

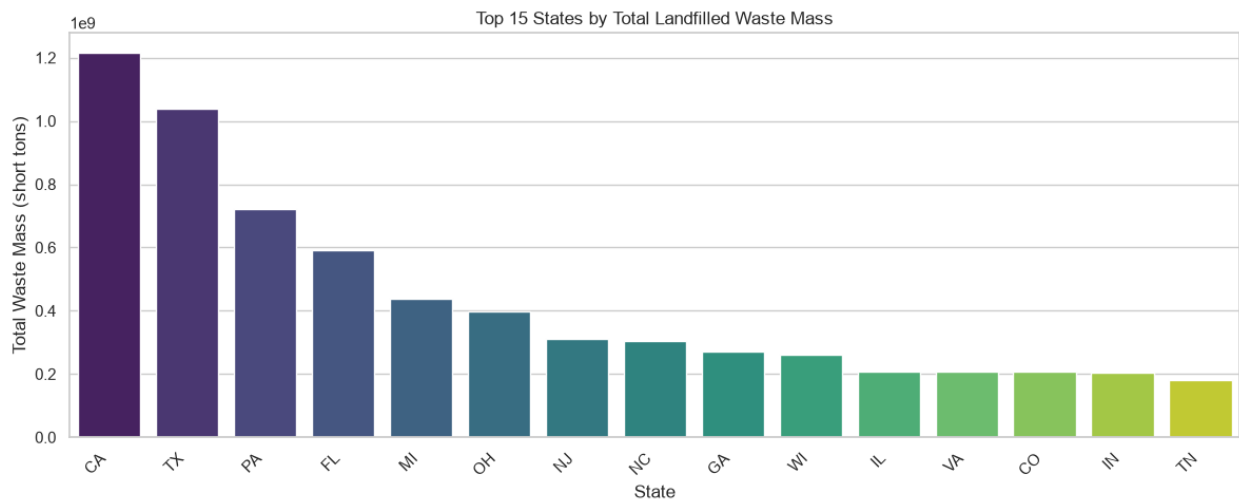


Figure 6 Top states by total landfilled waste mass.

5.7 Deployment Verification Results

The trained Linear Regression model, along with the label encoders, feature scaler, and target scaler were able to be successfully serialized and reloaded inside the Flask deployment

environment without having to retrain the model. Functional testing of the deployed application verified the correct rendering of the interface with the complete list of 45 recognized states and the six pre-computed sample forecasts and complete Table 1 metrics ranking.

The prediction endpoint was tried with both valid and invalid input conditions. The input validation logic that was put around the accepted state list worked as intended, as it returned a structured error response for a request made with an unrecognized state code, instead of an unhandled exception. An end-to-end pipeline from raw dataset to deployed forecast was confirmed to work by using a valid (two-letter) state code and forecast year in 1980 to 2035 range to get a complete forecast response, including the forecast value, model used, and entire historical series for that state.

The historical series also came back with the California forecast, adding 33 years of continuous yearly records, which further validates the aggregation and caching logic built into the `build_df_agg` function across the whole pre-processing pipeline – from raw data ingested from the Excel file, to the cached CSV file that is used at inference time. This continuity is essential for the lag-based feature reconstruction that is done in the `_predict` function because a gap in the yearly sequence would result in a lag value from an adjacent year.

Startup times were monitored during deployment tests, and the cached aggregate file gave the application the ability to start up without redoing the complete sequence of parsing and cleaning the Excel data, which verified that the caching described in Chapter 3 works as expected when the application is restarted repeatedly.

5.8 Summary

Based on the results presented in this chapter, Linear Regression, although it is a very simple structure compared to Random Forest, Gradient Boosting, XGBoost, and LSTM, was able to forecast the landfill waste mass most accurately on the processed state-year data with the R^2 value of 0.9861 as shown in Table 1. Ensemble tree-based models gave similar, but slightly lower predictions, and the LSTM model was the lowest of the five models, which is due to the relatively short, irregular length of historical sequences available per state, as discussed in Section 5.5. The results of the comparative visualizations depicted in the figures between 2 and 4 as well as the sample forecast verification described in section 5.6 validate that the chosen model and its accompanying feature engineering pipeline are indeed able to generate forecasts that align with the historical trends as observed. Results from deployment verification indicated that these findings are readily transferable to a working, dynamic forecasting tool, as per the goals outlined in Chapter 1.

Chapter 6: Conclusion and Future Work

6.1 Conclusion

The aim of this study was to identify which predictive modeling method would best predict landfill waste mass at a state-year level, and to see if a bottom-up model of landfill waste mass could be converted to a useful, deployable forecasting tool. The Landfilled Waste Composition Dataset was transformed into 846 state-year records, which account for 45 states and the time frame from 1938 to 2023. Categorical features were label encoded whereas the extreme values of the target were capped at the 99.9th percentile and an engineered lag1, lag2, lag3 and rolling three-year mean features were generated from the historical waste mass.

All train-test splits were consistent and used in the training and testing of five predictive models: Linear Regression, Random Forest, Gradient Boosting, XGBoost and a two-layered LSTM network using PyTorch. Comparative evaluation, summarized in Table 1, was done using RMSE, MAE, and R2, and the results showed that Linear Regression had the best performance with an R2 of 0.9861, an RMSE of ~1.05 million short tons, and an MAE of ~0.58 million short tons. There was very little difference in performance between Random Forest and Gradient Boosting, while XGBoost was slightly worse than those 2 and the LSTM network was the worst of the 5 candidates.

This result suggests a generally linear relationship between the historical lags and the current period waste mass for the seven-feature, log transformed representation of the state-year waste data employed in this study. The relatively short and uneven historical record length of several states (with average of about 18.8 years per state) seems to have hampered LSTM architecture in utilizing long-range temporal dependency which is discussed in detail in Chapter 5.

The chosen model - Linear Regression - along with its respective encoders and scalers, were successfully serialized and deployed using a Flask web app. Functional tests demonstrated that the deployed system successfully produces state-level forecasts, verifies state codes and year ranges, and provides full series of past observations with each forecast. A forecast for the State of California for 2027 yielded a value of 38,778,706 short tons, in line with the range for that state from 1990 to 2022.

The goals set in Chapter 1 have thus been fulfilled. The reproducible data processing pipeline was developed, five models were trained and thoroughly evaluated, the best-performing model was selected and explained, and a functioning forecasting interface was put in place, filling the gap between conceptual AI-powered waste management frameworks and empirically confirmed, deployable forecasting systems identified in Chapter 2.

6.2 Limitations

Some restrictions must be noted. This is because the data is not consistent for all states historically, and some states only have a few years on record, thus limiting extrapolation for these states in particular. The feature set is based on past mass of waste and time-dependent measures alone, with

no attempt to include other factors that might reasonably be expected to impact future volume of wastes to landfill, like population, economic activity or waste diversion policy. Although included for comparison purposes, the sequence length for the LSTM model was relatively small (three time steps), and this may have been insufficient to adequately demonstrate the capabilities of the recurrent architectures under more favorable data settings.

6.3 Future Work

This study could be continued in several directions. Forecasting accuracy may be improved by adding external socioeconomic factors, like the proportion of participating in recycling and diversion programs or state population estimates or gross domestic product. The addition of other more recent years of data, when they are available, would enable retraining and re-evaluation of all five models to verify if the ranking would remain the same over time.

Other sequence modeling options, such as attention-based models or hybrid statistical-neural models, might be considered, but may not be beneficial when longer historical sequences are available, compared to the LSTM configuration used in this study. The deployed application could also include uncertainty quantification (prediction intervals or quantile regression), giving decision makers some idea of the level of certainty around the forecast, which is more than a single point estimate.

Finally, the deployed application could be expanded to forecast at the facility level or waste-stream level instead of only at the state level, and could include periodic self-retraining based on new data on waste composition of landfills, thereby keeping the forecasting system in line with changing patterns of waste generation over time.

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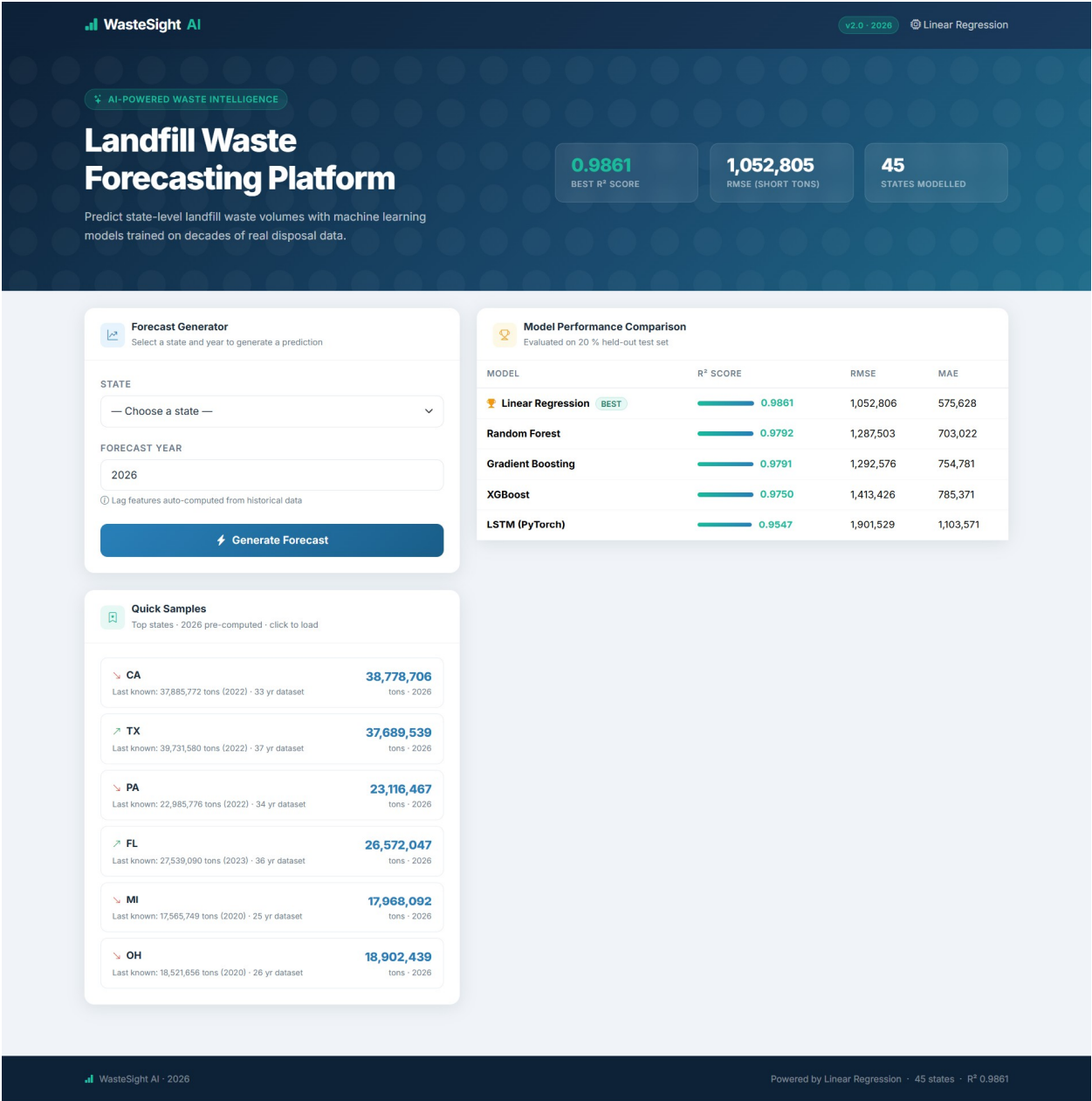
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Appendix: Web Application Demo



AI-POWERED WASTE INTELLIGENCE

Landfill Waste Forecasting Platform

Predict state-level landfill waste volumes with machine learning models trained on decades of real disposal data.

0.9861
BEST R² SCORE

1,052,805
RMSE (SHORT TONS)

45
STATES MODELLED

Forecast Generator

Select a state and year to generate a prediction

STATE

CO

FORECAST YEAR

2026

Lag features auto-computed from historical data

Generate Forecast

Quick Samples

Top states - 2026 pre-computed - click to load

CA

Last known: 37,885,772 tons (2022) - 33 yr dataset

38,778,706
tons - 2026

TX

Last known: 39,731,580 tons (2022) - 37 yr dataset

37,689,539
tons - 2026

PA

Last known: 22,985,776 tons (2022) - 34 yr dataset

23,116,467
tons - 2026

FL

Last known: 27,539,090 tons (2023) - 36 yr dataset

26,572,047
tons - 2026

MI

Last known: 17,565,749 tons (2020) - 25 yr dataset

17,968,092
tons - 2026

OH

Last known: 18,521,656 tons (2020) - 26 yr dataset

18,902,439
tons - 2026

Forecast Result

CO - 2026 forecast

Predicted Total Waste Mass

6,120,305

short tons

2026

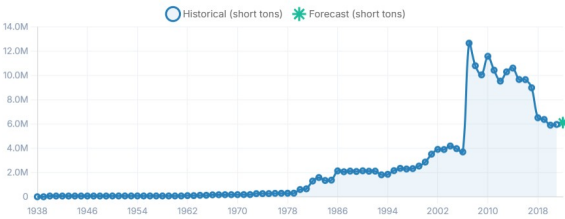
Linear Regression

INTERPRETATION

A 2.5% increase projected vs last recorded year (2021: 5,970,251 tons).

Historical Trend & Forecast

CO - 1938-2026



Model Performance Comparison

Evaluated on 20 % held-out test set

MODEL	R ² SCORE	RMSE	MAE
Linear Regression BEST	0.9861	1,052,806	575,628
Random Forest	0.9792	1,287,503	703,022
Gradient Boosting	0.9791	1,292,576	754,781
XGBoost	0.9750	1,413,426	785,371
LSTM (PyTorch)	0.9547	1,901,529	1,103,571